

Total No. of Questions : 8]

SEAT No. :

PC2808

[6352]-32

[Total No. of Pages :2

S.E. (Electronics & Computer Engg.)
ELECTRONICS CIRCUITS
(2019 Pattern) (Semester- III) (204202)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, & Q.7 or Q.8.
- 2) Figures to the right indicate full marks.

- Q1)** a) Draw practical voltage regulator using LM317 & justify each components. [5]
- b) Explain the basic principle of switching regulator with block diagram. [6]
- c) Explain low dropout regulator with neat diagram in detail [6]

OR

- Q2)** a) Compare linear regulator & SMPS. [5]
- b) Explain how current boosting is achieved in three terminal voltage regulator. [6]
- c) Draw and explain the block diagram of an adjustable three terminal regulator and justify function of each block. [6]

- Q3)** a) Draw block diagram of Op-Amp and Explain in detail [6]
- b) Define & Explain- [6]
- i) PSRR
 - ii) CMRR
 - iii) Slew Rate
- c) Draw the circuit diagram of dual input balance out differential amplifier & explain. [6]

OR

P.T.O

- Q4)** a) Explain current mirror circuit with neat diagram. [6]
 b) Draw the circuit diagram of level shifter and explain its necessity. [6]
 c) Define & Explain [6]
 i) I/P offset current
 ii) Gain Bandwidth product
 iii) I/P bias current

- Q5)** a) Draw & Explain schmitt trigger with characteristics. [6]
 b) Compare inverting & non inverting Amplifiers. [6]
 c) Draw the circuit diagram of differential amplifier & derive an expression for output voltage. [6]

OR

- Q6)** a) Draw the circuit diagram of square wave generator and explain with suitable diagram. [6]
 b) Draw the circuit diagram of an integrator and derive the expression for output voltage. [6]
 c) Draw the circuit diagram of instrumentation amplifier and derive the expression for output voltage. [6]

- Q7)** a) What is the operating principle of PLL & Explain its one application. [5]
 b) Explain the operation of successive approximation type ADC with neat block diagram. [6]
 c) Define [6]
 i) Free running frequency
 ii) Capture Range
 iii) Lock Range or PLL.

OR

- Q8)** a) What is the operating principle of DAC? Explain its one type. [5]
 b) Calculate the output frequency F_o , lock range ΔF_L and capture range ΔF_c of PLL IC565 if external resistor $R_1=12K\Omega$ external capacitor connected to pin no.9 $C_1=0.01 \mu F$, Filter capacitor C_f at pin no. 7= $10\mu f$. [6]
 c) Draw the block diagram of PLL & Explain in detail. [6]

